Queries based on Selection and Project operations:

XIT Database

1. List (ID, Name) of Students from Prog ID = 'BCS' and are having cpi > 7.5

$$\pi_{studid,\,name}\left(\sigma_{progid='BCS'\,AND\;cpi>7.5}(student)\right)$$

```
SELECT studid, name from student
       WHERE progid='BCS' AND cpi > 7.5;
```

2. List (ID, Name, cpi) of Students from Prog ID = 'BCS' or Prog ID = 'BIT' and are having cpi > 7.5

$$\pi_{studid,\,name,\,cpi}\left(\sigma_{(progid='BCS'\,OR\;progid='BIT')\;AND\;cpi>7.5}(student)\right)$$

```
SELECT studid, name, cpi from student
       WHERE (progid='BCS' or progid='BIT') AND cpi > 7.5;
```

Company Database

1. List all employee of DNO=4

$$\sigma_{dno=4}\,(employee)$$

```
SELECT * FROM employee WHERE dno=4;
```

2. List employees having salary > 30000

$$\sigma_{salary\,>\,30000}(employee)$$

```
SELECT * FROM employee WHERE salary > 30000;
```

3. List (ENO, Name, Salary) for all employees

$$\pi_{eno,ename,\,salary}(employee)$$

```
SELECT eno, ename, salary FROM employee;
```

4. List (ENO, Name, Salary) for all employees of DNO=4

$\pi_{eno,ename,\,salary}(\sigma_{dno=4}\,(employee))$

```
SELECT eno, ename, salary FROM employee WHERE dno=4;
```

5. List (ENO, Name, Salary) of all Female Employees

$\pi_{eno,ename,\,salary}(\sigma_{gender='F'}(employee))$

```
SELECT eno, ename, salary FROM employee WHERE gender='F';
```

6. List eno, name, salary of employees who are supervised by eno=105

$\pi_{eno,ename,\,salary}(\sigma_{super_eno=105}(employee))$

```
SELECT eno, ename, salary FROM employee WHERE super_eno=105;
```

7. List (ENO, Name, Salary) having salary >= 10000 and <= 30000

$\pi_{eno,ename,\,salary}(\sigma_{salary\ >=\ 10000\ and\ <=\ 30000}(employee))$

```
SELECT eno, ename, salary FROM employee
       WHERE salary >= 10000 and <= 30000;
```

8. List employees either working dno=4 and salary > 25000 or working dno=5 and salary > 30000.

$\sigma_{(dno=4\ and\ salary\ >\ 25000)\ or\ (dno=5\ and\ salary\ >\ 30000)}(employee)$

```
SELECT eno, ename, salary FROM employee
   WHERE (dno=4 and salary > 25000) or (dno=5 and salary > 30000);
```

9. List employees who are older than 50 years

$\sigma_{(year(currentdate-dob)\geq\ 50)}(employee)$

```
select * FROM employee where extract(year from age(dob)) >= 50;
```

Queries based on JOIN operations:

XIT Database

1. List (prog-name, dept-name) of the program offered at XIT

$$\pi_{pname,dname}(program * department)$$

```
select pname, dname from program
     natural join department;
```

2. List students who are studying in CS department

$$r1 \leftarrow student \overset{\bowtie}{\underset{progid=pid}{}} program * department$$
$$\pi_{s.*}(\sigma_{dname='CS'}(r1))$$

```
select s.* from student as s join program as p on progid=pid
    natural join department
    where dname='CS';
```

3. List (stud-name, prog-name) of all students

$$\pi_{name,pname}\left(student \overset{\bowtie}{\underset{progid=pid}{}} program\right)$$

```
select name, pname from student join program on progid=pid;
```

4. List student name, program name of students having cpi > 7.5

$$\pi_{name,pname}\left(\sigma_{cpi>7.5}\left(student \overset{\bowtie}{\underset{progid=pid}{}} program\right)\right)$$

```
select name, pname from student join program on progid=pid
    where cpi > 7.5;
```

5. What name of the department in which student id = 123 is studying?

$$r1 \leftarrow student \overset{\bowtie}{\underset{progid=pid}{}} program * department$$
$$\pi_{dname}(\sigma_{sid='123'}(r1))$$

```
select dname from student as s join program as p on progid=pid
    natural join department
    where studid='123';
```

6. List ID and Name of students of BTech(CS) and BTech(ECE)

$$\pi_{sid,name}(\sigma_{progid='BCS'\,OR\,progid='BCE'}(student))$$

```
select studid, name from student
    where progid='BCS' or progid='BCE';
```

Company Database

1. List ENO, Name, DName of all employees

$$\pi_{eno,ename,dname}(employee * department)$$

```
select eno,ename,dname from employee
     natural join department where dno = 4;
```

2. List ename, salary of all Female Supervisors

$$\pi_{s.ename,s.salary}\left(\sigma_{s.gender='F'}\left(employee\ e \overset{\bowtie}{\underset{e.super_eno=s.eno}{}} employee\ s\right)\right)$$

```
select distinct s.ename, s.salary from employee as e
    join employee as s on e.super_eno=s.eno
    where s.gender='F';
```

3. List ename, salary of all Female Managers

$$\pi_{ename,salary}\left(\sigma_{gender='F'}\left(employee \overset{\bowtie}{\underset{mgr_eno=eno}{}} department\right)\right)$$

```
select ename, salary from employee
    join department on mgr_eno=eno
    where gender='F';
```

4. List employees having salary > their manager (manager of their department)

$$r1 \leftarrow employee\ e * department \overset{\bowtie}{\underset{mgr_eno=m.eno}{}} employee\ m$$
$$\pi_{e.eno,e.ename}\left(\sigma_{e.salary>m.salary}(r1)\right)$$

```
select e.eno, e.ename from employee e natural join department
    join employee as m on mgr_eno=m.eno
    where e.salary > m.salary;
```

5. List Employee that are working on projects monitored by DNO=4

$$r1 \leftarrow employee\ e * works_on\ w \overset{\bowtie}{\underset{w.pno=p.pno}{}} project\ p$$
$$\pi_{e.eno,e.ename}\left(\sigma_{p.dno=4}(r1)\right)$$

```
select e.eno, e.ename from employee e natural join works_on w
      join project p on w.pno=p.pno
    where p.dno=4;
```

6. List Employee Name, Supervisor Name (if any)

$$\pi_{e.ename,s.ename}\left(employee\ e \overset{LEFT\ \bowtie}{\underset{e.super_eno=s.eno}{}} employee\ s\right)$$

```
select e.ename, s.ename from employee as e
    left join employee as s on e.super_eno=s.eno;
```

7. List employees (eno, name) that work on projects managed by department 'Research'.

$$r1 \leftarrow employee\ e * works_{on} w \overset{\bowtie}{\underset{w.pno\ =p.pno}{}} project\ p$$
$$\pi_{e.eno,e.ename}\left(\sigma_{dname='Research'}(r1 \overset{\bowtie}{\underset{p.dno\ =d.dno}{}} department\ d)\right)$$

```
select e.eno, e.ename from employee e natural join works_on w
      join project p on w.pno=p.pno join department d on p.dno=d.dno
    where dname='Research';
```

Queries based on SET operations:

Company Database

1. List Eno of non-managers, i.e., they are not manager of any department

$$\pi_{eno}(employee) - \pi_{mgr_eno}(department)$$

```
select eno from employee
except
select mgr_eno from department;
```

2. List Eno, Ename, and Salary of non-managers, i.e., they are not managers of any department

$$r1 \leftarrow \pi_{eno}(employee) - \pi_{mgr_eno}(department)$$
$$result \leftarrow \pi_{eno,ename,salary}(employee * r1)$$

```
select eno, ename, salary from employee natural join
    (select eno from employee
     except
    select mgr_eno from department) as r;
```

OR

$$\pi_{eno,ename,salary}\left(employee \overset{semi-diff}{\underset{eno=mgr_eno}{}} department\ d)\right)$$

```
select eno, ename, salary from employee where eno
      not in (select mgr_eno from department);
```

3. Find out Employees (eno, ename, salary) that are directly (works for) for dno=4 or indirectly associated with (work on projects managed by) dno=4

$$r1 \leftarrow \pi_{eno}(\sigma_{dno=4}(employee)) \cup \pi_{eno}(\sigma_{dno=4} works_on * project)$$
$$result \leftarrow \pi_{eno,ename,salary}(employee * r1)$$

```
select eno, ename, salary from employee natural join
    (select eno from employee where dno=4
     union
    select eno from works_on natural join project where dno=4) as r;
```

4. List ENO, ENAME of employees who work on at least one project.

$$\pi_{eno,ename}\left(employee\ e \frac{semi-intersection}{e.eno=w.eno} works_on\ w)\right)$$

```
select eno, ename from employee where eno in
    (select eno from works_on);
```

5. Compute employees (eno, ename, salary) that are supervisors (supervises some employees) but not managers (that are manager of any department)

$$r1 \leftarrow \pi_{super_eno}(employee) - \pi_{mgr_eno}(department)$$
$$result \leftarrow \pi_{eno,ename,salary}\left(employee\ e \frac{\bowtie}{e.eno=r1.super_eno} r1\right)$$

```
select eno, ename from employee e join
    (select distinct super_eno from employee
    except
    select mgr_eno from department
   ) as r
  on r.super_eno=e.eno;
```

6. Compute employees (eno, ename, salary) that are supervisors or Managers but not both

$$r1 \leftarrow \pi_{super_eno}(employee)$$
$$r2 \leftarrow \pi_{mgr_eno}(department)$$
$$r3 \leftarrow (r \cup r2) - (r \cap r2)$$
$$result \leftarrow \pi_{eno,ename,salary}\left(employee\ e \frac{\bowtie}{e.eno=r3.super_eno} r3\right)$$

```
select eno, ename, salary from employee e join
    ((select distinct super_eno from employee
    union
    select mgr_eno from department)
except
(select distinct super_eno from employee
    intersect
    select mgr_eno from department
   )) as r
  on r.super_eno=e.eno;
```